# HW5 Econ 144

Mason Lee

2025-11-29

```r
library(readr)
library(lattice)
library(foreign)
library(MASS)
library(car)
require(stats)
require(stats4)
library(KernSmooth)
library(fastICA)
library(cluster)
library(leaps)
library(mgcv)
library(rpart)
library(pan)
library(mgcv)
library(DAAG)
library("TTR")
library(tis)
require("datasets")
require(graphics)
library(forecast)
require(astsa)
library(RColorBrewer)
library(plotrix)
library(tsibble)
library(tseries)
library(ggplot2)
library(tidyr)
library(USgas)
library(fpp3)
```

## Chapter 12

### 12.3

```r
data123 <- read_csv("data123.csv")
```

```
## Rows: 169 Columns: 4
```

```
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): obs
## dbl (3): CPI_Germany, CPI_US, EX_dollar_euro
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
us <- ts(data123$CPI_US, start=c(1999,1), frequency=12)
gr <- ts(data123$CPI_Germany, start=c(1999,1), frequency=12)
ex <- ts(data123$EX_dollar_euro, start=c(1999,1), frequency=12)

gr_usd <- gr * ex
diff_ppp <- us - gr_usd

adf.test(diff_ppp)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  diff_ppp
## Dickey-Fuller = -2.5342, Lag order = 5, p-value = 0.3533
## alternative hypothesis: stationary
```

```
adf.test(us)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  us
## Dickey-Fuller = -2.8167, Lag order = 5, p-value = 0.2354
## alternative hypothesis: stationary
```

```
adf.test(gr_usd)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  gr_usd
## Dickey-Fuller = -2.53, Lag order = 5, p-value = 0.355
## alternative hypothesis: stationary
```

The high p values in the above tests prove that we can reject the null hypothesis, and that all three relationships are non-stationary. This means that the two series are not cointegrated and that PPP does not hold.

## 12.4

```r
data124 <- read_csv("data124.csv")
```

```
## Rows: 224 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl  (2): cpi, nominal
## date (1): date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
cpi <- ts(data124$cpi, start=c(1957,1), frequency=12)
i <- ts(data124$nominal, start=c(1957,1), frequency=12)

pi <- diff(log(cpi)) * 100
rt <- window(i, start=start(pi)) - pi

adf.test(pi)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  pi
## Dickey-Fuller = -2.4771, Lag order = 6, p-value = 0.3757
## alternative hypothesis: stationary
```

```r
adf.test(i)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  i
## Dickey-Fuller = -2.9123, Lag order = 6, p-value = 0.1929
## alternative hypothesis: stationary
```

```r
adf.test(rt)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  rt
## Dickey-Fuller = -2.8677, Lag order = 6, p-value = 0.2117
## alternative hypothesis: stationary
```

Due to the high p values in the above three adf tests, inflation, normal rate, and real rate are all non stationary. This means that he Fisher relationship does not produce a stationary real rate. Overall, we do not have strong evidence that the Fisher equation will hold here.

## 12.5

```r
data125 <- read_csv("data125.csv")
```

```
## Rows: 714 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): obs
## dbl (2): TB3Month, TB10Year
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
data125 <- data125[complete.cases(data125), ]

short <- ts(data125$TB3Month, start=c(1954,4), frequency=12)
long <- ts(data125$TB10Year, start=c(1954,4), frequency=12)

adf.test(short)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  short
## Dickey-Fuller = -2.2379, Lag order = 8, p-value = 0.4776
## alternative hypothesis: stationary
```

```r
adf.test(long)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  long
## Dickey-Fuller = -1.4983, Lag order = 8, p-value = 0.7907
## alternative hypothesis: stationary
```

```r
spread <- long - short
adf.test(spread)
```

```
## Warning in adf.test(spread): p-value smaller than printed p-value
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  spread
## Dickey-Fuller = -4.5083, Lag order = 8, p-value = 0.01
## alternative hypothesis: stationary
```

```r
adf.test(long - short)
```

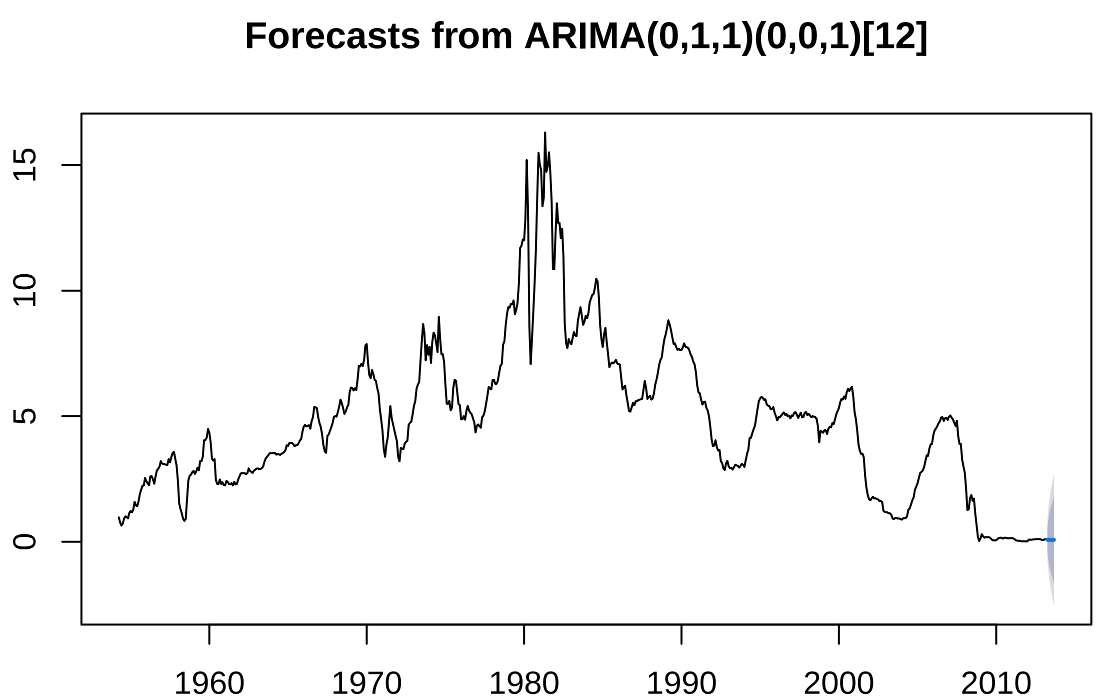
```
## Warning in adf.test(long - short): p-value smaller than printed p-value
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  long - short
## Dickey-Fuller = -4.5083, Lag order = 8, p-value = 0.01
## alternative hypothesis: stationary
```

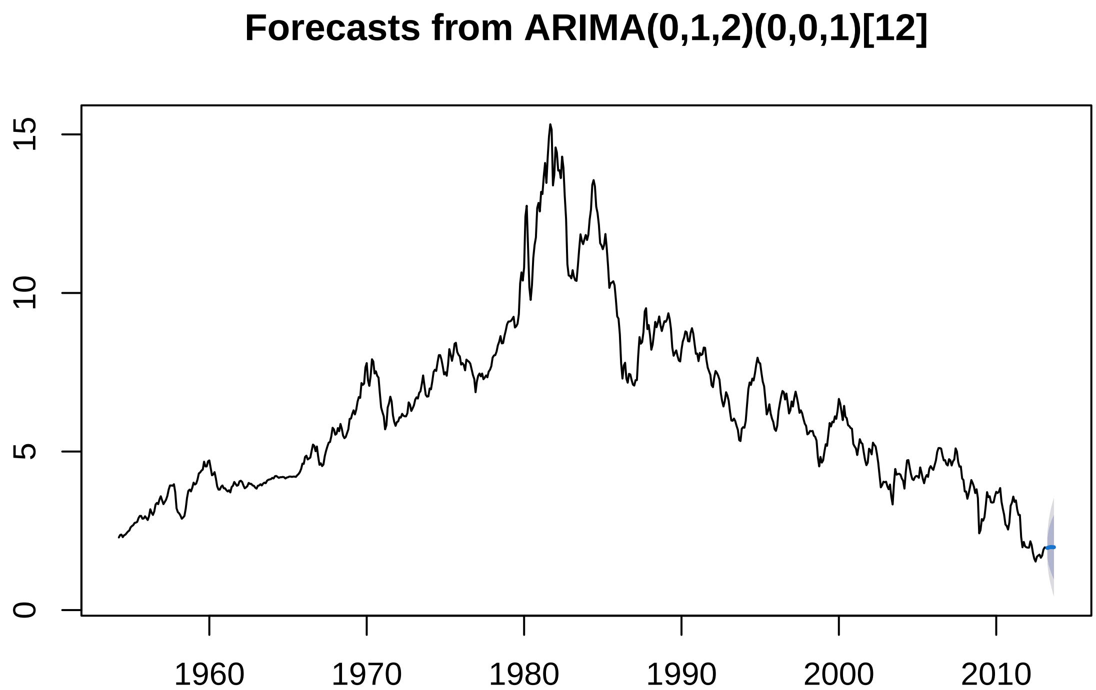
The short and long rates are both non stationary (high p values), but the spreads have very low p values, meaning we can reject the null hypothesis. This means the short and long rates are cointegrated despite neither rates being stationary.

### 12.6

```
fit_short <- auto.arima(short)
fit_long <- auto.arima(long)

a <- forecast(fit_short, h=6)
b <- forecast(fit_long, h=6)

plot(a)
```




```
plot(b)
```




```
a
```

```
##          Point Forecast      Lo 80     Hi 80      Lo 95     Hi 95
## Apr 2013     0.07889728 -0.4283020 0.5860966 -0.6967971 0.8545917
## May 2013     0.07786278 -0.8130736 0.9687991 -1.2847068 1.4404323
## Jun 2013     0.07796482 -1.0754146 1.2313442 -1.6859767 1.8419064
## Jul 2013     0.07647429 -1.2898348 1.4427833 -2.0131151 2.1660637
## Aug 2013     0.07686578 -1.4733977 1.6271293 -2.2940576 2.4477892
## Sep 2013     0.07574616 -1.6388480 1.7903404 -2.5464994 2.6979917
```

```
b
```

```
##          Point Forecast    Lo 80    Hi 80     Lo 95    Hi 95
## Apr 2013       1.958733 1.631135 2.286330 1.4577159 2.459749
## May 2013       1.974459 1.410585 2.538333 1.1120884 2.836830
## Jun 2013       1.984878 1.277940 2.691816 0.9037102 3.066046
## Jul 2013       1.989849 1.164277 2.815420 0.7272460 3.252452
## Aug 2013       1.983777 1.054597 2.912958 0.5627188 3.404836
## Sep 2013       1.982556 0.960214 3.004898 0.4190188 3.546093
```

### 12.7

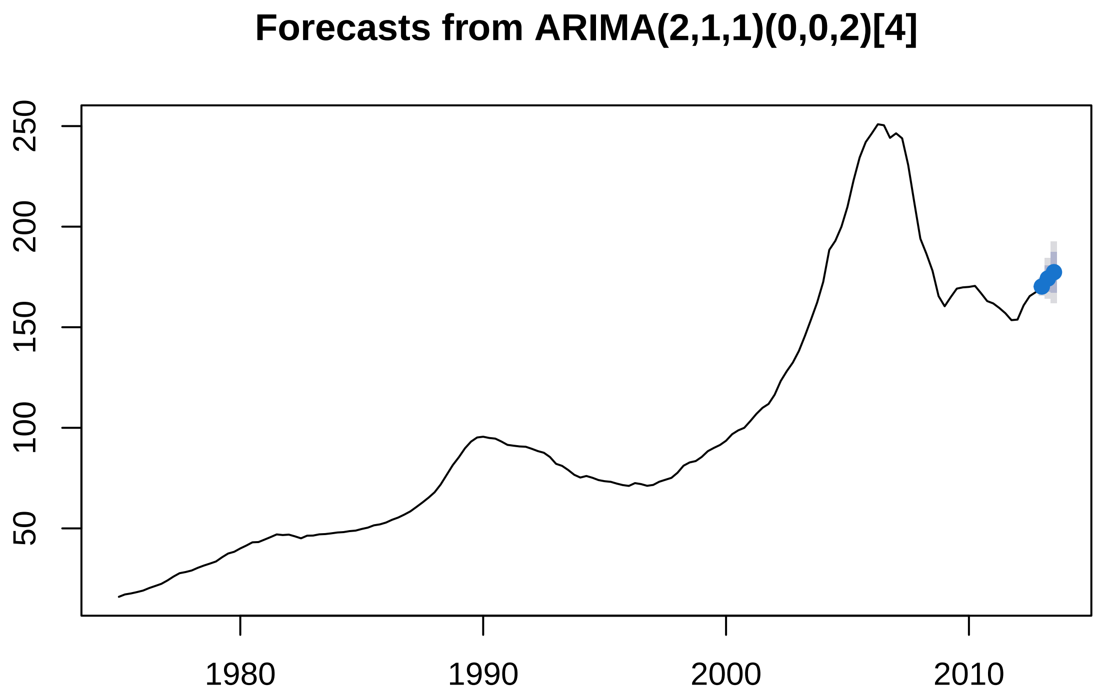
```r
data127 <- read_csv("data127.csv")
```

```
## Rows: 152 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): obs
## dbl (2): ILA, IRV
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

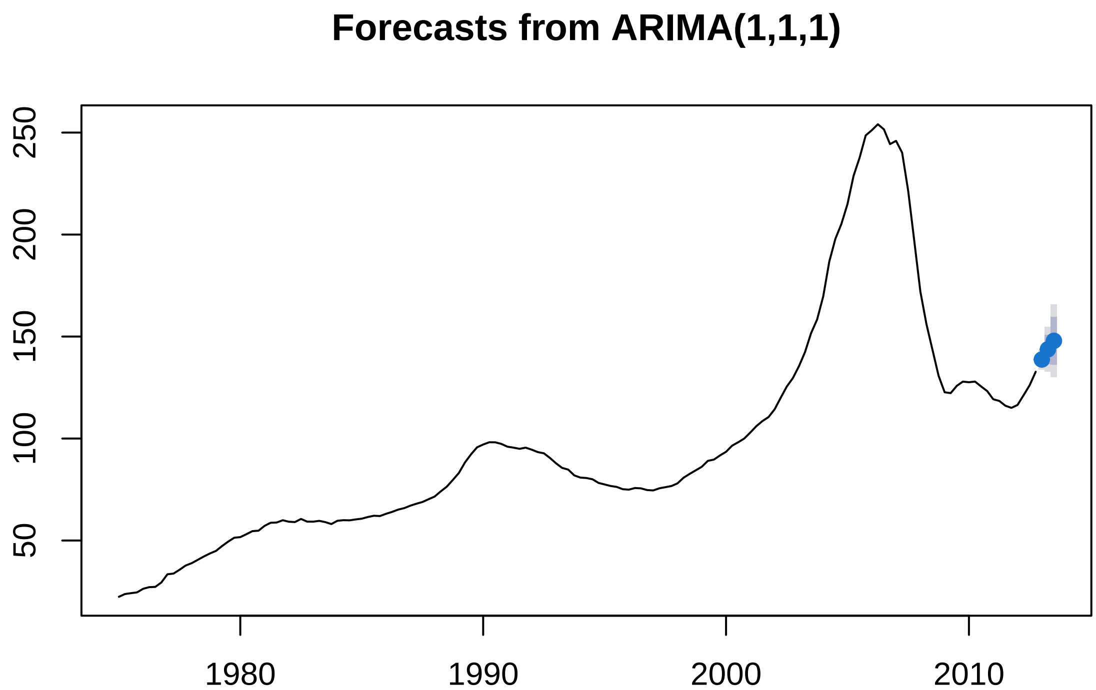
```r
data127 <- data127[complete.cases(data127), ]

la <- ts(data127$ILA, start=c(1975,1), frequency=4)
riv <- ts(data127$IRV, start=c(1975,1), frequency=4)

adf.test(la - riv)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  la - riv
## Dickey-Fuller = -3.2763, Lag order = 5, p-value = 0.07782
## alternative hypothesis: stationary
```

```r
fit_la <- auto.arima(la)
fit_riv <- auto.arima(riv)

c <- forecast(fit_la, h=3)
d <- forecast(fit_riv, h=3)

plot(c)
```


Forecasts from ARIMA(2,1,1)(0,0,2)[4]
250
200
150
100
50
1980
1990
2000
2010

```r
plot(d)
```




```
c
```

```
##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## 2013 Q1       170.2537 167.2799 173.2276 165.7056 174.8019
## 2013 Q2       174.2376 167.6584 180.8167 164.1757 184.2995
## 2013 Q3       177.3682 167.3079 187.4285 161.9823 192.7541
```

```
d
```

```
##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## 2013 Q1       138.7435 135.5563 141.9308 133.8691 143.6180
## 2013 Q2       143.7405 136.5337 150.9472 132.7187 154.7623
## 2013 Q3       147.9135 136.2269 159.6000 130.0405 165.7865
```

The p value fromn the adf test is 0.0778, which is close to stationary but not quite low enough to reject the null hypothesis (0.05). If this is the only test we can use, then we would say that LA and Riverside home prices are not stationary, but more exploration is needed to say for certain what is happening between these two datasets. There is a possible cointegrated relationship.